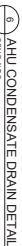
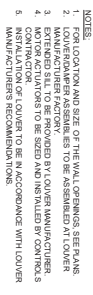


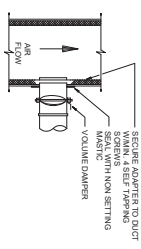
0 2 4
SCALE 1/4" = 1'-0"

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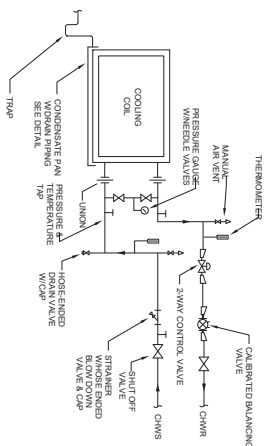


8 TYPICAL PIPE HANGER DETAIL
SCALE: 1" = 1'-0"

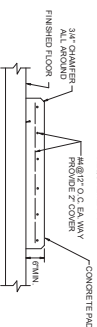
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4 ROUND DUCT TAKE-OFF DETAIL



6 CHILLED WATER COIL PIPING DIAGRAM (2-WAY VALVE)
- SCALE: 1"=8'



8 CONCRETE EQUIPMENT PAD DETAIL
S.C.A.E. N.T.S.

110024 B-1 Project/NIER 012 97N00-01 297-100 29 CAD Conf in SRM-2 MECHANICAL DETAILS.dwg

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AIR TERMINAL BOX SCHEDULE										
MARK	MANUFACTURER (MODEL)	NET L/D (INCHES)	REGULATOR SET CTM	MIN. CTM	HEATING (BTU)	ELECTRIC HEATING KW	VOLTS /PH /HZ	CONTROL S	MAX. KW	REMARKS
VAW-1	TRANE /VOCF	14	2770	870	410	12	480 /3 /60	DDC	30	SEE NOTES
VAW-2	TRANE /VOCF	16	360	290	137	4	277 /1 /60	DDC	30	SEE NOTES
VAW-3	TRANE /VOCF	6	360	360	137	4	277 /1 /60	DDC	30	SEE NOTES
VAW-4	TRANE /VOCF	6	320	125	5.1	1.5	277 /1 /60	DDC	30	SEE NOTES
VAW-5	TRANE /VOCF	6	460	190	8.5	2.5	277 /1 /60	DDC	30	SEE NOTES
VAW-6	TRANE /VOCF	6	430	175	8.5	2.5	277 /1 /60	DDC	30	SEE NOTES
VAW-7	TRANE /VOCF	4	130	870	3.4	1.0	277 /1 /60	DDC	30	SEE NOTES
VAW-8	TRANE /VOCF	8	400	400	17.1	5.0	480 /3 /60	DDC	30	SEE NOTES
VAW-9	TRANE /VOCF	4	85	85	3.4	1.0	277 /1 /60	DDC	30	SEE NOTES

PAN SCHEDULE													
MASK	QTY	MANUFACTURER MODEL	CFM	7.5 IN W.C.I.	NOTICE	HP	ELECTRIC IN W.C.I.	TYPE	DRIVE				
FE-1	1	GREENHECK CASH-904	165	0.5	1725	1/10	115	1	90	CENTRIFUGAL, SERIAL	DIRECT	RESTROOMS	REF NOTES 1, 2, 3
FE-2	1	GREENHECK CASH-904	400	0.5	1725	1/15	115	1	90	CENTRIFUGAL, SERIAL	DIRECT	RESTROOMS	REF NOTES 1, 2, 3
FE-3	1	GREENHECK CASH-904	300	0.4	1725	1/14	115	1	90	CENTRIFUGAL, SERIAL	DEL.T	RESTROOMS	REF NOTES 1, 2, 4
FE-4	1	GREENHECK CASH-904	300	0.4	1725	1/10	115	1	90	CENTRIFUGAL, SERIAL	DEL.T	RESTROOMS	REF NOTES 1, 2, 4
FE-5	1	GREENHECK CASH-904	200	0.4	1725	1/10	115	1	90	CENTRIFUGAL, SERIAL	DEL.T	RESTROOMS	REF NOTES 1, 2, 4, 5
FE-6	1	SHAWCRAFT D800P	30	-	-	1/2	120	1	60	NINE DOLT	DIRECT	MECHANICAL 11A	REF NOTES 1, 2, 3, 6
FE-7	1	SHAWCRAFT D800P	30	-	-	1/2	120	1	60	NINE DOLT	DIRECT	MECHANICAL 11A	REF NOTES 1, 2, 3, 6

PACKAGED AIR COOLED CHILLER SCHEDULE												
		MECHANICAL				ELECTRICAL						
MARK	MANUFACTURER MODEL	NOMINAL CAPACITY (TONS)	OPERATING WATER FLOW (GPM)	MANUFACTURER TYP	EVAP. LAT. TYP	COND. WAT. P. (FT)	VOLTS	Ø	HZ.	TOTAL KW	REMARKS	
CH-1	THANE (CDA-2)	20	35.6	4.3	59/44	16.1/17.22	0.35	400	3	60	25.9	SEE NOTES

SPLIT SYSTEM AIR CONDITIONING UNITS									
MARK	NIDROS UNIT TYPE	REFRIGERANT TYPE	FPAI	MOA		WEIGHT (LBS)	BASED ON MANUFACTURER MODEL	REMARKS	
				COOLING BTU/H	ELECTRICAL (V/A/Hz)				(INDOOR / OUTDOOR)
A/C1+ACDU1	WALL MOUNTED	R410A	530	24	208 V / 60	1 / 19	53 / 193	MITSUBISHI PAK-AIR/PDV-XANNA	SEE NOTES

RUN CONDITIONS - SCHEDULED:
THE UNIT SHALL RUN TO MAINTAIN

2

- * A 75°F (ADJ.) COOLING SETPOINT
 - * A 70°F (ADJ.) HEATING SETPOINT.
- NIGHT SETBACK FUNCTIONALITY SHALL NOT BE INITIALLY UTILIZED. HOWEVER, THE CAPABILITY TO PROVIDE NIGHT SETBACKS SHALL BE INCLUDED.
- ALARMS SHALL BE PROVIDED AS FOLLOWS:

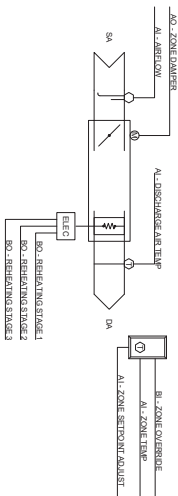
- HIGH ZONE TEMP: IF THE ZONE TEMPERATURE IS GREATER THAN THE COOLING SETPOINT BY A USER-DEFINABLE AMOUNT (ADJ.).
- LOW ZONE TEMP: IF THE ZONE TEMPERATURE IS LESS THAN THE HEATING SETPOINT BY A USER-DEFINABLE AMOUNT (ADJ.).

ZONE SETPOINT ADJUST:

THE OCCUPANT SHALL BE ABLE TO ADJUST THE ZONE TEMPERATURE

- WHEN ZONE TEMPERATURE IS GREATER THAN ITS COOLING SETPOINT, THE ZONE DAMPER SHALL MODULATE BETWEEN THE MINIMUM OCCUPIED AIRFLOW (AOJ) AND THE MAXIMUM COOLING AIRFLOW (AOJ) UNTIL THE ZONE IS SATISFIED.
- WHEN THE ZONE TEMPERATURE IS BETWEEN THE COOLING SETPOINT AND THE HEATING SETPOINT, THE ZONE DAMPER SHALL MAINTAIN THE MINIMUM REQUIRED ZONE VENTILATION

- WHEN ZONE TEMPERATURE IS LESS THAN ITS HEATING SETPOINT, THE CONTROLLER SHALL ENABLE HEATING TO MAINTAIN THE ZONE TEMPERATURE AT ITS HEATING SETPOINT



1	VARIABLE AIR VOLUME TERMINAL UNIT (VAV-1 THRU VAV-9)
-	SCALE N.T.S.

EXHAUST FAN - ON/OFF (EF-1 THRU EF-3)

RUN CONDITIONS - INTERLOCKED:
THE FANS SHALL BE INTERLOCKED TO RUN WHENEVER A HANDLING UNIT APU-1 IS IN OCCUPIED MODE UNLESS SHUTDOWN ON SAFETIES.

THE FAN SHALL HAVE A USER DEFERRABLE (VFD) / INVERTER RUN TIME.

EXHAUST AIR DAMPER

THE EXHAUST AIR DAMPER SHALL OPEN ANYTIME THE UNIT RUNS AND SHALL CLOSE ANYTIME THE UNIT STOPS. THE EXHAUST AIR DAMPER SHALL CLOSE 30 SEC (ADJ.) AFTER THE FAN STOPS.

- DAMPER FAILURE: COMMANDED OPEN, BUT THE STATUS IS CLOSED.
- DAMPER IN HAND: COMMANDED CLOSED, BUT THE STATUS IS OPEN.

DAMPER STATUS:
THE FAN SHALL BE ENABLED AFTER THE DAMPER STATUS HAS

- DAMPER FAILURE: COMMAND OPEN, BUT THE STATUS IS

- DAMPER IN HAND: COMMANDED CLOSED, BUT THE STATUS IS OPEN.

- FAN FAILURE: COMMAND ON, BUT THE STATUS IS OFF.
- FAN IN HAND: COMMAND OFF, BUT THE STATUS IS ON.
- FAN RUNTIME EXCEEDED: FAN STATUS RUNTIME EXCEEDS A USER DEFINABLE LIMIT (ADU).

UNOCCUPIED:

- WHEN THE ZONE IS UNOCCUPIED THE ZONE DAMPER SHALL CLOSE TO ITS MINIMUM (UNOCCUPIED AIRFLOW ADJ.)
- WHEN THE ZONE TEMPERATURE IS GREATER THAN ITS COOLING SETPOINT THE ZONE DAMPER SHALL MODULATE BETWEEN THE MINIMUM (UNOCCUPIED AIRFLOW ADJ.) AND THE MAXIMUM (COOLING AIRFLOW ADJ.) UNTIL THE ZONE IS SATISFIED
- WHEN ZONE TEMPERATURE IS LESS THAN ITS UNOCCUPIED HEATING SETPOINT, THE CONTROLLER SHALL Bypass HEATING TO MAINTAIN THE ZONE TEMPERATURE AT THE SETPOINT.

ELECTRIC REHEATING STAGES:

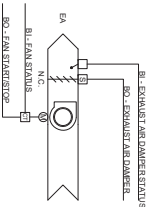
REHEATING STAGE 1				X			
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STAGE THE REHEATING TO MAINTAIN ITS SETPOINT TO PREVENT SHORT CYCLING, THERE SHALL BE A USER DEFINABLE (ADJ.) IDEAL STAGES, AND EACH STAGE SHALL HAVE A USER DEFINABLE (ADJ.) MINIMUM RUNTIME.

- OUTSIDE AIR TEMPERATURE IS LESS THAN 46°F (ADJ.)
 - AND THE ZONE TEMPERATURE IS BELOW SETPOINT
 - AND SUFFICIENT AIR FLOW IS PROVIDED
- DISCHARGE AIR TEMPERATURE**
- THE CONTROLLER SHALL MONITOR THE DISCHARGE AIR TEMPERATURE.
- ALARMS SHALL BE PROVIDED AS FOLLOWS
- HIGH DISCHARGE AIR TEMP. IF THE DISCHARGE AIR TEMPERATURE IS GREATER THAN 127°F (ADJ.)
 - LOW DISCHARGE AIR TEMP. IF THE DISCHARGE AIR TEMPERATURE IS LESS THAN 46°F (ADJ.)

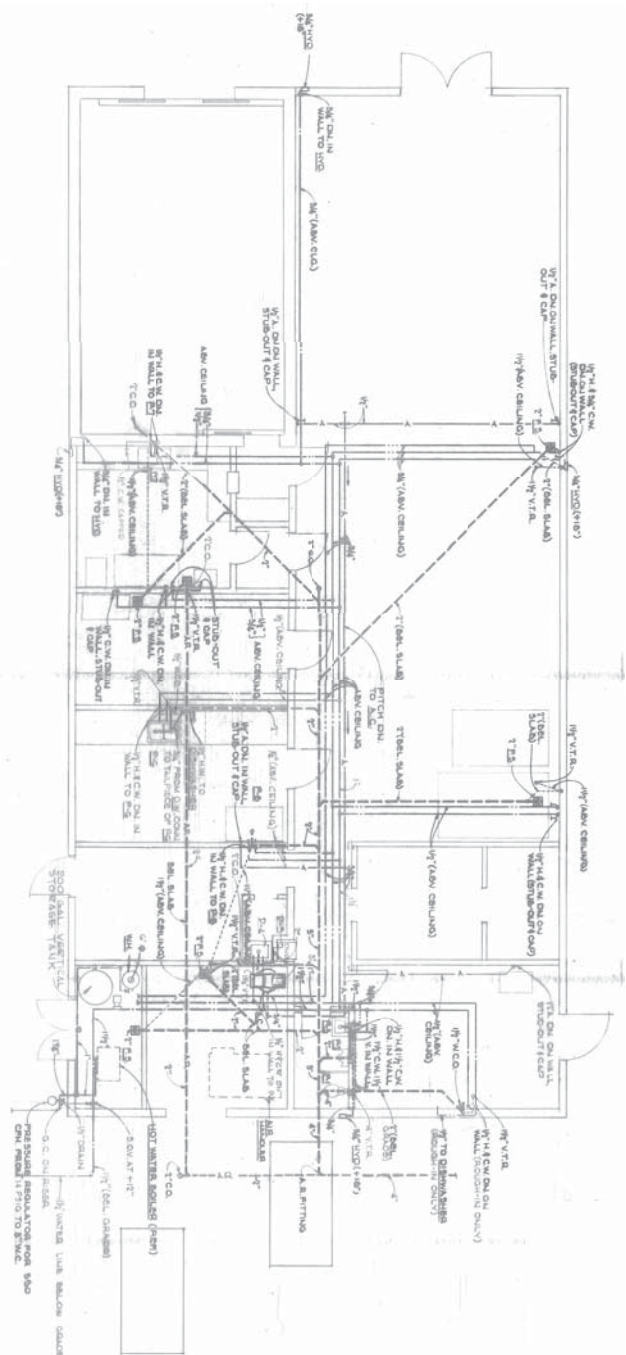
POINT NAME	HARDWARE POINTS				SOFTWARE POINTS				ELEMENT TYPE
	A1	A0	B1	B0	A1	B1	SCHED/INTERVAL		
ARTFLOW	X				X		GRAPHIC		
DISCHARGE ART TEMP	X				X				
ZONE SETPOINT ADJUST	X				X				
ZONE TEMP	X				X				
ZONE DAMPER			X		X				
REHEATING STAGE 1				X	X				
REHEATING STAGE 2					X				
REHEATING STAGE 3				X	X				
ARTFLOW SETPOINT					X				
COOL AND SETPOINT					X				
HEATING SETPOINT				X	X				
HEATING MODE					X				
SCHEDULE						X			
THRU DISCHARGE ART TEMP							X		
ZONE 1 DISCHARGE ART TEMP							X		
ZONE 2 DISCHARGE ART TEMP							X		
ZONE 3 DISCHARGE ART TEMP							X		
LOW ZONE TEMP							X		
LOW ZONE TIME							X		
TOTALS	4	1	1	3	3	1	12	4	
TOTAL HARDWARE (B1) TOTAL SOFTWARE (C1)									
								13	

EXHAUST FAN - (NO/OF (E3 - THREE-3))									
POINT NAME	HARDWARE POINTS				SOFTWARE POINTS				STATION OR OFFICE
	A/	B/	BO	AV	BV	SCHEDULING/MAINT	TESTING		
EXHAUST FAN DAMPER STATUS		X				X		X	
FAN STATUS		X				X		X	
EXHAUST FAN DAMPER			X			X		X	
FAN START/STOP				X			X	X	
EXHAUST FAN DAMPER IN FAULT						X		X	
EXHAUST FAN DAMPER IN HAND						X		X	
FAN FAILURE						X		X	
FAN RUN						X		X	
FAN BLADE EXCEEDED						X		X	
TOTALS	0	0	2	2	0	0	4	5	4
TOTAL HARDWARE (A)									
TOTAL SOFTWARE (B)									



2 EXHAUST FAN - ON/OFF (EF-1 THRU EF-3)

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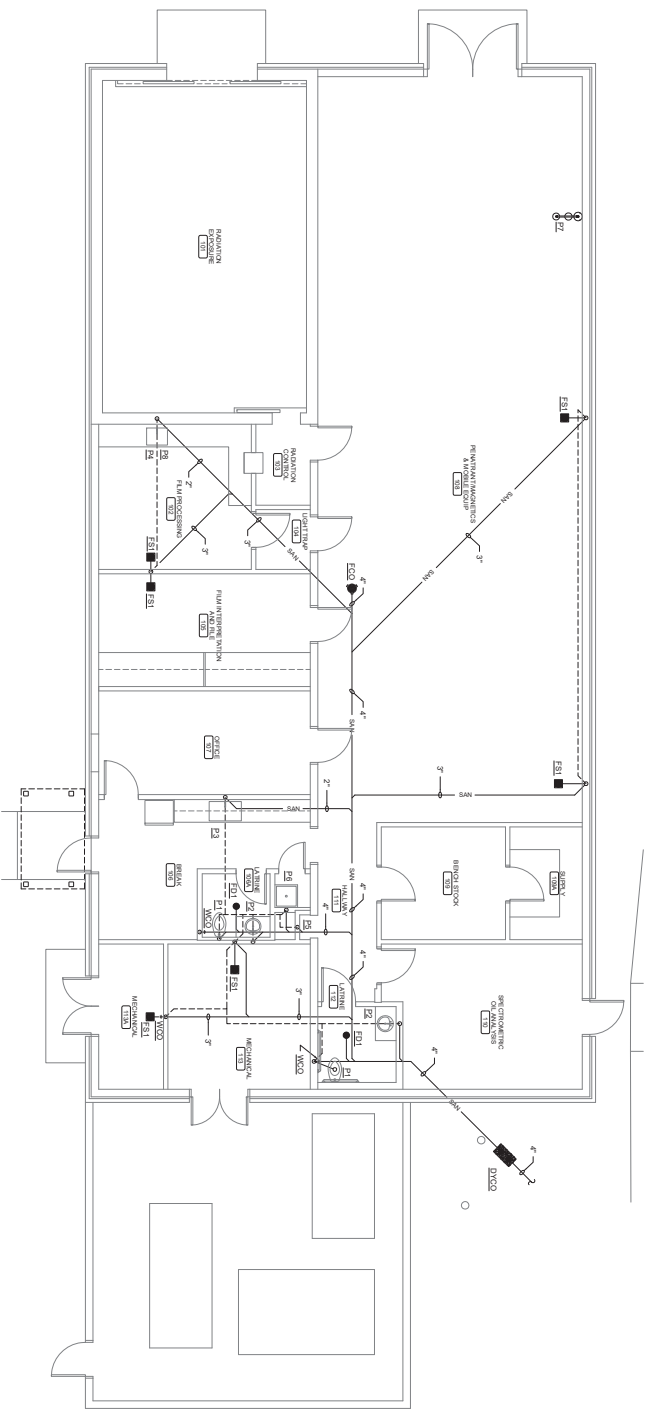
- DEMOLITION NOTES:
1. ALL EXISTING WATER PIPING ABOVE AND BELOW EQUIPMENT AND ACCESSORIES TO BE REMOVED TO CONSTRUCTION SPAN, TURNING AROUND BACK FLOW PREVENTION WITHIN MECHANICAL ROOMS TO CONSTRUCTION SPAN.
 2. ALL EXISTING SANITARY SEWER AND VENT PIPING ABOVE AND BELOW SLAB TO BE REMOVED.
 3. ALL EXISTING COMPRESSED AIR PIPING, EQUIPMENT, AND ACCESSORIES TO REMAIN.
 4. ALL EXISTING PLUMBING FIXTURES TO BE REMOVED.



SCALE: 1/4" = 1'-0"



PLUMBING DEMOLITION PLAN	11, 11
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BUILDING 5406
SANITARY SEWER FLOOR PLAN

1
SCALE 1/4" = 1'-0"

N

SCALE 1:40,000

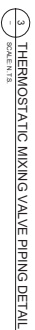
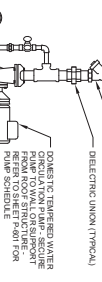
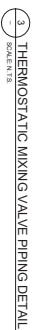


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WATER HEATER SCHEDULE										
MODE	AREA SERVED AND APPROXIMATE LOCATION	MINIMUM STORAGE CAPACITY (GAL.)	MINIMUM DISCHARGE TEMP °F	ELECTRICAL			NATURAL GAS		FLOW RATE (GPM)	
				TOTAL KW	VOLTA GE PHASE	NUMBER OF LINES	C/H	BURNING IN W.C.		FLUE TO P. SIZE
MH-1	MECHANICAL ROOM H15	40	140°	55°	6	480V 3	-	-	-	MANUFACTURER MODEL BRANDS/WHITE GOODS
		27					C/H	BURNING IN W.C.	FLUE TO P. 100° F	

NOTES:
1. PIPING TO AND FROM EACH WATER HEATER AND HOT WATER STORAGE TANK SHALL BE ROUTED HORIZONTALLY AND DOWNWARD A MINIMUM OF 2 FEET BEFORE TURNING IN AN UPWARD DIRECTION.

EXPANSION TANK SCHEDULE

CIRCULATING PUMP SCHEDULE							
MARK	MANUFACTURER / MODEL (SERIES / PART NO.)	TD IN FT	ELECTRICAL		SERVING	LOCATION	REMARKS
			HP	V			
OP-1	4	40	125	115	1	00	100% RETURN
							MECH RM
							STAINLESS STEEL, HOUSING AND BANG CONTROLS

[illegible]

NON-DESTRUCTIVE INSPECTION (NDI) LAB: FIRE PROTECTION

GENERAL NOTES

FIRE PROTECTION SCOPE OF WORK:
INSTALL A NEW AUTOMATIC SPRINKLER SYSTEM AND REPLACE THE EXISTING FIRE ALARM SYSTEM WITH NEW.

FIRE SPRINKLER GENERAL NOTES:

- A. INSTALL AN AUTOMATIC SPRINKLER SYSTEM IN THE EXISTING BUILDING IN ACCORDANCE WITH 2016 NFPA 13 AND UFC 3-600.0-1.
- B. PERFORM OR OBTAIN A CURRENT WATER FLOW TEST FOR HYDRAULIC CALCULATIONS TO BE PREPARED WITH SHOP DRAWINGS. WATER FLOW DATA PROVIDED AS PART OF THIS SUBMITTAL IS ONLY FOR A GENERAL REFERENCE.
- C. INSTALL PIPE HANGERS AND SEISMIC BRACING ON ALL NEW SPRINKLER PIPING IN ACCORDANCE WITH NFPA 13.
- D. PROVIDE COMPLETE SHOP DRAWINGS AND HYDRAULIC CALCULATIONS FOR REVIEW PRIOR TO INSTALLATION.

FIRE ALARM GENERAL NOTES:

- A. REPLACE THE EXISTING FIRE ALARM SYSTEM IN THE EXISTING BUILDING WITH A NEW ADDRESSABLE FIRE ALARM / MASS NOTIFICATION SYSTEM.
- B. THE FIRE ALARM SYSTEM SHALL BE INSTALLED IN ACCORDANCE WITH 2016 NFPA 72, IFC 3400-01, AND IFC 3400-1.
- C. THE EXISTING ALARM PANEL SHALL BE REPLACED WITH A NEW ALARM PANEL.
- D. ALL NEW WIRING, DEVICES, AND FIRE ALARM CONTROL PANELS SHALL BE INSTALLED IN ACCORDANCE WITH THE LATEST EDITION OF THE NATIONAL ELECTRICAL CODE (NEC).
- E. ALL NEW WIRING SHALL BE RUN IN CONDUIT. EXISTING FIRE ALARM CONDUIT MAY BE RE-USED AT THE DISCRETION OF THE CONTRACTOR, IF RE-USED EXISTING CONDUIT SHALL BE TAGGED EVERY TEN FEET WITH A RED STRIKE AT LEAST 1" INCH IN WIDTH.
- F. THE FIRE ALARM CONTROL PANEL WILL ALSO SERVE AS THE MASS NOTIFICATION CONTROL PANEL. A SEPARATE LOCAL OPERATING CONSOLE (LOO) SHALL BE PROVIDED ADJACENT TO THE FIRE ALARM / MASS NOTIFICATION CONTROL PANEL. THE CONTROLS OF THE LOO SHALL BE ACCESSIBLE WITHOUT THE USE OF A KEY OR PASSCODE.
- G. THE FIRE ALARM REPORTING SYSTEM SHALL BE COMPATIBLE WITH THE MONACO RADIO SYSTEM.
- H. AN EXTERIOR, WALL MOUNTED ANTENNA MUST BE PROVIDED FOR THE MONACO RADIO SYSTEM.
- I. PROVIDE COMPLETE SHOP DRAWINGS AND HYDRAULIC CALCULATIONS FOR REVIEW PRIOR TO INSTALLATION.

APPLICABLE CODES

- UFC 1-200-01 GENERAL BUILDING REQUIREMENTS, 20 JUNE 2016
- UFC 3-310-04 SEISMIC DESIGN FOR BUILDINGS, 15 JUNE 2013
- UFC 3-600-01 FIRE PROTECTION ENGINEERING FOR FACILITIES, WITH CHANGE 1, 28 NOVEMBER 2016
- UFC 4-021-01 DESIGN AND O&M: MASS NOTIFICATION SYSTEMS, WITH CHANGE 1, JANUARY 2010
- INTERNATIONAL BUILDING CODE (IBC), 2015 (AS REFERENCED BY UFC)

THE NATIONAL FIRE PROTECTION ASSOCIATION (NFPA) CODES AND STANDARDS (AS REFERENCES BY THE UFC), INCLUDING, BUT NOT LIMITED TO:

- NFPA 10, STANDARD FOR PORTABLE FIRE EXTINGUISHERS, 2013
- NFPA 13, STANDARD FOR THE INSTALLATION OF SPRINKLER SYSTEMS, 2016
- NFPA 70, NATIONAL ELECTRICAL CODE, 2017
- NFPA 72, NATIONAL FIRE ALARM AND SIGNALING CODE, 2016
- NFPA 101, LIFE SAFETY CODE, 2015

WHERE CONFLICTS EXIST BETWEEN THE UFC AND OTHER INDUSTRY CODES AND STANDARDS, THE UFC TAKES PRECEDENCE.

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